

LUCAS CRUZ

Senior Data Engineer | Gen. AI Automation Engineer

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Florianópolis, Santa Catarina, Brazil · Requires visa sponsorship

PROFESSIONAL SUMMARY

Senior Data Engineer and Gen. AI Automation Engineer with 5+ years of experience building production data pipelines, cloud lakehouses, AI agents, and LLM/n8n automation across IoT, fintech, agribusiness, and automotive intelligence. Proven track record delivering end-to-end data platforms, governed analytics, conversational AI workflows, and customer-facing product development. Currently extending this work toward production AI platforms (RAG, agents, evaluation, MCP, observability) over enterprise data. Fluent in English (C1).

TECHNICAL SKILLS

Gen. AI & LLMs	LLM Integration, AI Agents (OpenAI, Claude, Gemini...), RAG, Prompt Engineering, Chatbot Development
Data Engineering & Orchestration	Python, SQL, Apache Spark (PySpark), Pandas, FastAPI, NestJS, Next.js, dbt, Trino, REST APIs, Web Scraping, Playwright, Parquet, Apache Airflow, n8n, Celery, Redis, Meta Cloud API
Cloud & Lakehouses	Amazon Web Service (AWS), Microsoft Azure, Google Cloud (GCP), Oracle Cloud (OCI), Snowflake, Databricks, Amazon Redshift, Delta Lake, Oracle Autonomous Database
Relational Databases & BI	PostgreSQL, SQL Server, MySQL, Redis, Power BI, Qlik Sense, Apache Superset, Looker
Security & Data Governance	LGPD/GDPR Compliance, Data Privacy, IAM, RBAC, Data Masking, Encryption (Rest/Transit), Audit Logging, RFID/NFC Tagging, Firewall
DevOps & Infrastructure	Docker, Kubernetes, Azure Bicep, Azure Container Apps, Git, CI/CD Pipelines, Linux, Bash Scripting
Languages	Portuguese (Native) · English (C1 - Advanced) · Spanish (B1 - Intermediate)

WORK EXPERIENCE

Gen. AI Engineer — TK Technologies | Curitiba, PR - Brazil (Hybrid) *Feb 2026 - Present · Contract*

- Architected OrbitAI, an omnichannel AI CRM platform (WhatsApp, Instagram, Facebook, Telegram, and Email) with role-based data governance, Meta/Google Ads tracking, and LLM-powered auto-replies, deploying to Azure Container Apps with Bicep IaC.
- Engineered a multi-source web scraping pipeline in Python targeting 10+ automotive parts websites, extracting pricing and product metadata at scale to deliver competitive market intelligence.
- Architected AI-powered n8n automation workflows (ETL, RPA, and chatbot agents) by integrating LLMs (Claude AI, Google AI Studio) and GCP APIs (Sheets, Drive) to enrich datasets and reduce manual analysis work.
- Developed a conversational AI chatbot that empowers business users to query and refine pricing datasets via natural language, reducing time-to-insight for non-technical stakeholders.
- Coordinated Notion-based delivery sprints, customer-facing demos, product training, sales support, and post-launch support; contribute to the TK web presence and ongoing CRM product development and maintenance.

Data Engineer — Spacecom Monitoramento | Curitiba, PR - Brazil (On-site) *Apr 2023 - Jul 2024*

- Designed and maintained an enterprise Big Data Lakehouse on AWS S3 leveraging Medallion Architecture (Bronze/Silver/Gold) alongside Delta Lake, Parquet, PySpark, and Trino to support heavy analytics workloads.
- Supported a high-availability IoT platform scaling to 95,000+ devices, efficiently processing hundreds of millions of telemetry records per day.

- Managed electronic security tag (RFID/NFC) data for government court systems, ensuring strict LGPD/GDPR compliance through robust data privacy controls, access governance, and secure handling of sensitive judicial data.
- Orchestrated and optimised ETL pipelines with Apache Airflow, reducing failure rates by ~30%; utilised Docker on AWS EC2 to cut deployment issues by an additional ~25%.
- Remediated inconsistencies in pipeline steps to boost downstream analytics accuracy by ~40%, successfully delivering reliable datasets to Qlik Sense and Apache Superset dashboards.

Data Engineer Consultant — wDiscover | Curitiba, PR - Brazil (Remote) *Jun 2022 - May 2024*

- Delivered end-to-end data projects across the fintech, agribusiness, and operations sectors by integrating high-volume payment providers (PagSeguro, Stone, Cielo), John Deere IoT APIs, and REST APIs.
- Designed dimensional Data Warehouses on SQL Server and Oracle Autonomous Database; automated complex ETL workflows with Airflow on Docker/Oracle Cloud to reduce manual effort by ~35%.
- Built dynamic Power BI dashboards to track critical KPIs, financial metrics, and operational performance for diverse client domains.

Data Analyst — Itaete Group | Curitiba, PR - Brazil (Hybrid) *Jun 2020 - Feb 2023*

- Designed a SQL Server Data Warehouse and robust Python/SQL pipelines using Pandas to cleanse, transform, and enrich data from IoT devices, PostgreSQL, APIs, and CSV sources.
- Built Power BI dashboards and automated legacy Excel/VBA reporting workflows, slashing data processing time by ~50% and significantly reducing manual operational workloads.
- Developed internal systems for inventory management and financial automation, streamlining invoice generation and automated weekly report distribution.

NOTABLE PROJECTS

OrbitAI — Omnichannel AI CRM · *NestJS, Next.js, PostgreSQL, LLMs, Meta & Google Ads APIs, Azure Container Apps, Docker* · *TK Technologies - Jun 2026 - Present*

- Built OrbitAI, a secure omnichannel CRM integrating WhatsApp, Instagram, Facebook, Telegram, and Email into a unified inbox with role-based access control (RBAC).
- Layered Meta/Google Ads performance data and LLMs (Claude/Google AI Studio) for auto-replies, outbound messaging assistance, and automated campaign analysis.

Car Parts Price Searcher · *Python, FastAPI, Celery, Playwright, Redis, PostgreSQL, n8n, Azure, Bicep* · *TK Technologies - Feb 2026 - Present*

- Built a car-parts price searcher across 10+ sites, with parallel public-price and internal-stock lookup, auditable reporting, and notifications from containerised Azure services.

IoT Data Lakehouse on AWS - Medallion Architecture · *AWS S3, Apache Spark, Delta Lake, Airflow, Trino, Apache Superset* · *Client Project - Feb 2024*

- Designed a Bronze/Silver/Gold lakehouse on AWS S3 ingesting from APIs and databases. Virtualised the Gold layer with Trino and surfaced governed datasets through BI dashboards.

EDUCATION

Bachelor of Systems Analysis and Development — Dom Bosco University, Curitiba, Brazil · Jun 2023

Majors: Information Systems · Business Analytics · Data Analysis · Database Systems · Software Engineering

LICENSES & CERTIFICATIONS

- AI for Writing and Communicating — Google (Credential ID: HE8YE5IS7WVW) · Jul 2026
- AI for Research and Insights — Google (Credential ID: 4PZDF3XQOF55) · Jul 2026
- AI for Brainstorming and Planning — Google (Credential ID: 7SFW2HGOKIU5) · Jul 2026
- AI Fundamentals — Google (Credential ID: UKBGM0AZUIS2) · Jul 2026